

VICTORIA TUCK

vtuck [at] engineering.upenn.edu

CURRENT RESEARCH

Autonomous systems have moved from single agents in controlled environments to many interacting agents woven into our most critical infrastructure, where uncertainty, competing incentives, and constant change make dependable operation both essential and enormously difficult. In my research, **I develop algorithmic frameworks with theoretical guarantees for multi-agent cyber-physical systems** in these settings. Specifically, I design for **safety under uncertainty**, **social alignment among self-interested agents**, and **efficient, sustained coordination**. I leverage tools from online learning, optimization, mechanism design, and formal methods and demonstrate my methods in applications including next-generation air mobility and multi-robot systems. More broadly, **my long-term vision is to make the design of large-scale multi-agent systems as principled and routine as modern software engineering**.

ACADEMIC APPOINTMENTS

University of Pennsylvania

Postdoctoral Researcher

2025 – present

Advisor: George J. Pappas

EDUCATION

University of California, Berkeley

Ph.D., Electrical Engineering and Computer Sciences

2025

Advisors: Shankar Sastry, Sanjit Seshia

Georgia Institute of Technology

M.S., Electrical and Computer Engineering

2018

B.S., Electrical Engineering, exchange at the Technical University of Munich

2017

PUBLICATIONS

* denotes equal contribution

Google Scholar

Journal Articles

- [1] Maria G. Mendoza, **Victoria M. Tuck**, Chinmay Maheshwari, S. Shankar Sastry. “Resilient Decentralized Ergodic Coverage for Scalable Multi-Robot Systems in Unknown Time-Varying Environments.” Under review, *IEEE Robotics and Automation Letters (RA-L)* (2026). [arXiv:2604.04280](https://arxiv.org/abs/2604.04280)
- [2] Chinmay Maheshwari*, Maria G. Mendoza*, **Victoria M. Tuck***, Pan-Yang Su, Victor Qin, Hamsa Balakrishnan, Sanjit Seshia, Shankar Sastry. “Privacy Preserving Mechanisms for Coordinating Airspace Usage in Advanced Air Mobility.” *ACM J. Auton. Transport. Syst.* (2025).

Conference Papers

- [3] Matthew Low, Jasmine J. Aloor, **Victoria M. Tuck**, Pierluigi Nuzzo, Jason J. Choi. “Time-To-Reach Separation and Safety Filtering for Safe, Fair, and Efficient Multi-Agent Coordination.” *To appear, 64th IEEE Conference on Decision and Control (CDC)*, 2026. [arXiv:2605.20625](https://arxiv.org/abs/2605.20625).
- [4] Pan-Yang Su*, Chinmay Maheshwari*, **Victoria M. Tuck**, S. Shankar Sastry. “Incentive-Compatible Vertiport Reservation in Advanced Air Mobility: An Auction-Based Approach.” *63rd IEEE Conference on Decision and Control (CDC)*, Milan, Italy, 2024.
- [5] **Victoria M. Tuck**, Pei-Wei Chen, Georgios Fainekos, Bardh Hoxha, Hideki Okamoto, S. Shankar Sastry, Sanjit A. Seshia. “SMT-Based Dynamic Multi-Robot Task Allocation.” *NASA Formal Methods Symposium (NFM)*, 2024.

- [6] **Victoria M. Tuck**, Yash V. Pant, Sanjit A. Seshia, S. Shankar Sastry. “DEC-LOS-RRT: Decentralized Path Planning for Multi-robot Systems with Line-of-sight Constrained Communication.” *IEEE Conference on Control Technology and Applications (CCTA)*, 2021.

Workshops

- [7] Maria G. Mendoza, **Victoria M. Tuck**, Chinmay Maheshwari, S. Shankar Sastry. “Decentralized Ergodic Coverage Control in Unknown Time-Varying Environments.” *AAMAS 2026 Workshop on Autonomous Robots and Multirobot Systems*, 2026.
- [8] **Victoria M. Tuck**, Hardik Parwana, Pei-Wei Chen, Georgios Fainekos, Bardh Hoxha, Hideki Okamoto, S. Shankar Sastry, Sanjit A. Seshia. “MRTA-Sim: A Modular, Open-World Simulator for Benchmarking Multi-Robot Systems.” *RSS 2025 Workshop on Scalable and Resilient Multi-Robot Systems*, 2025. (poster)
- [9] **Victoria M. Tuck**. “Coordinating Multi-Agent Systems at Different Levels of Interaction.” *CPS Week 2025 PhD Forum*, 2025. (poster)
- [10] **Victoria M. Tuck**, Maïke Schwammberger, Martin Fränzle, Sanjit A. Seshia, S. Shankar Sastry. “Safe and Performant Driving in Occluded Environments with Information-Aware Game Theory.” *NSF PIRE Workshop on Societal-Scale Cyber-Physical Systems*, 2023. (poster)

Technical Reports

- [11] J. J. Parish, D. Whitten, M. Gandhi, G. Jares, L. Hood, **V. Tuck**, K. Williams, R. Schlossman, D. Potter, G. Bennett, D. Kouri, B. Nicholson. “Rapid, Robust, Autonomous Trajectory Generation for Hypersonic Vehicles.” SAND2019-12442, Sandia National Laboratories, 2019.

INVITED TALKS

- 2026 RGSO-CARS Seminar Series, University of Delaware. (*Upcoming, title TBD*)
- 2026 “From Local Coordination to System-Level Design of Trustworthy Multi-Agent Systems.” ACC Workshop on Autonomy in Transportation.
- 2026 “From Local Coordination to System-Level Strategies: Designing Multi-Agent Autonomy Across Scales.” N. Matni’s group, University of Pennsylvania.
- 2025 “Coordinating Multi-Agent Systems at Different Levels of Interaction.” Autonomy Talks.
- 2025 “Market Mechanisms for Advanced Air Mobility.” NASA Ames. (*with C. Maheshwari, M. G. Mendoza, P.-Y. Su*)
- 2024 “Mechanism Design for Advanced Air Mobility.” INFORMS (*invited session*).

AWARDS

NextProf Nexus Acceptance	2024
Berkeley EECS Outstanding Teaching Assistant Award	2024
Berkeley Outstanding Graduate Student Instructor Award	2021
Berkeley EECS Excellence Award	2019

TEACHING EXPERIENCE

University of California, Berkeley

Fall 2023, 2024 Embedded Systems (UG/Grad) — Guest Lecturer

Fall 2020, 2023 Embedded Systems (UG/Grad) — Graduate Student Instructor

Georgia Institute of Technology

Fall 2018 Graduate Linear Controls (Grad) — Teaching Assistant

Spring 2018 Undergraduate Microelectronics Lab (UG) — Teaching Assistant

2014 – 2017 MATLAB for Engineers (UG) — Teaching Assistant

RESEARCH MENTORING

- Maria G. Mendoza (UC Berkeley) — Younger PhD mentee; joint ACM JATS (2025) publication and RA-L submission under review.
- Danil Kuznetsov (UCLA) — Transfer to Excellence summer researcher; presented at SACNAS (2024).

SOFTWARE

MRTA-Sim

Modular, open-world simulator for benchmarking multi-robot task allocation, planning, and control.

github.com

CPSGrader

Contributor: Automatic grading and feedback-generation tool for cyber-physical systems labs.

cpsgrader.org

INDUSTRY EXPERIENCE

Toyota Research Institute of North America (TRINA)

Cyber-Physical Systems Research Resident

2024

- Built testbed simulator for dynamic multi-agent task allocation, planning, and control in complex indoor environments
- Researched strategies for deadlock avoidance and locally centralized multi-agent deconfliction

Zipline

Automation Intern

2022

- Implemented method to adjust drone paths on-line to manage in-air density for safe operation at higher densities
- Enabled a 30% increase in current in-air capacity, with support for future increase

Sandia National Laboratories

Controls Engineering Intern — Navigation, Guidance, and Control

2019

- Analyzed impact of stochastic control on trajectory planning for hypersonics

ACADEMIC SERVICE

IEEE Robotics and Automation Letters (RA-L), Reviewer

2026

Artificial Intelligence (Elsevier), Reviewer

2026

IEEE Conference on Decision and Control (CDC), Reviewer

2026

International Conference on Learning Representations (ICLR) VerifAI Workshop, Reviewer

2025, 2026

UC Berkeley EE Faculty Hiring Committee, Member

2025

IEEE Conference on Decision and Control (CDC), Session Co-Chair

2024

UC Berkeley EE Graduate Admissions, Application Reviewer

2021, 2024

IEEE/ACM International Conference on Embedded Software (EMSOFT), Reviewer

2023

IEEE International Conference on Robotics and Automation, Reviewer

2023

OUTREACH

Bringing science education to local K-12 schools

Bay Area Scientists in Schools (BASIS), Facilitator

2019 – 2025

Be A Scientist, Mentor

2021

Supporting women and girls in STEM

Girl Scouts Engineering Day (Berkeley WiCSE), Organizer

2021 – 2023

Berkeley WiCSE, Outreach Chair

2020 – 2023

Expand Your Horizons, Workshop Leader

2019 – 2020

Girls In Engineering, Volunteer

2020

Mentoring and Supporting under-represented and first-generation students
Transfer to Excellence (TTE), Mentor

2023

PROFESSIONAL AFFILIATIONS

IEEE Control Systems Society • IEEE Robotics and Automation Society • IEEE Women in Engineering (2019 – Present)

REFERENCES

Shankar Sastry — UC Berkeley

sastry@eecs.berkeley.edu

Sanjit Seshia — UC Berkeley

sseshia@eecs.berkeley.edu

Georgios Fainekos — Toyota of North America

georgios.fainekos@toyota.com

Hamsa Balakrishnan — MIT

hamsa@mit.edu